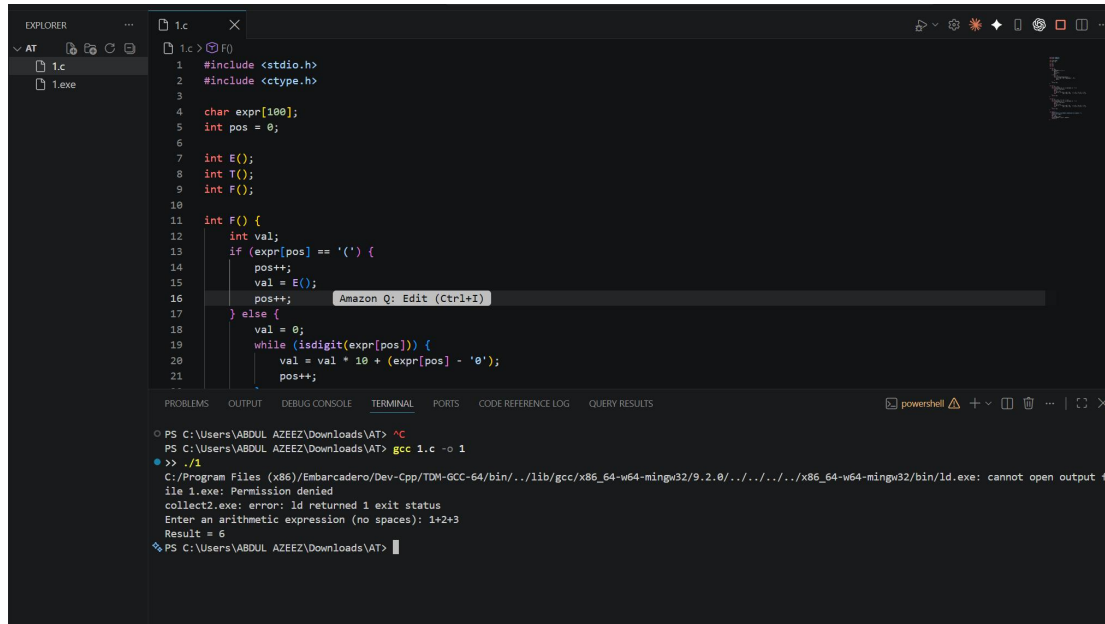


CSA1402 COMPILER DESIGN(CO3_AT1)

ALNAJEEB 192424325



```
1 #include <stdio.h>
2 #include <ctype.h>
3
4 char expr[100];
5 int pos = 0;
6
7 int E();
8 int T();
9 int F();
10
11 int F() {
12     int val;
13     if (expr[pos] == '(') {
14         pos++;
15         val = E();
16         pos++;
17     } else {
18         val = 0;
19         while (isdigit(expr[pos])) {
20             val = val * 10 + (expr[pos] - '0');
21             pos++;
22         }
23     }
24 }
```

PS C:\Users\ABDUL AZEEZ\Downloads\AT> gcc 1.c -o 1

PS C:\Users\ABDUL AZEEZ\Downloads\AT> ./1

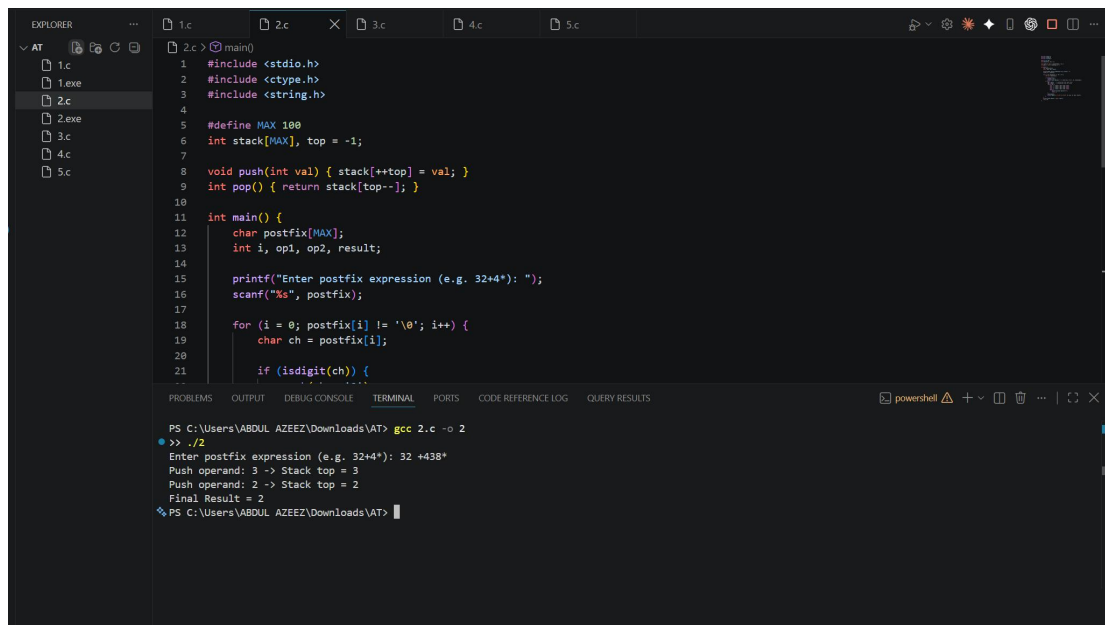
C:\Program Files (x86)\Embarcadero\Dev-Cpp\TDM-GCC-64\bin\..\lib\gcc\x86_64-w64-mingw32\9.2.0\..\..\x86_64-w64-mingw32\bin\ld.exe: cannot open output file 1.exe: Permission denied

collect2.exe: error: ld returned 1 exit status

Enter an arithmetic expression (no spaces): 1+2+3

Result = 6

PS C:\Users\ABDUL AZEEZ\Downloads\AT>



```
1 #include <stdio.h>
2 #include <ctype.h>
3 #include <string.h>
4
5 #define MAX 100
6 int stack[MAX], top = -1;
7
8 void push(int val) { stack[++top] = val; }
9 int pop() { return stack[top--]; }
10
11 int main() {
12     char postfix[MAX];
13     int i, op1, op2, result;
14
15     printf("Enter postfix expression (e.g. 32+4*): ");
16     scanf("%s", postfix);
17
18     for (i = 0; postfix[i] != '\0'; i++) {
19         char ch = postfix[i];
20
21         if (isdigit(ch)) {
22             push(ch - '0');
23         } else if (ch == '+' || ch == '*' || ch == '/' || ch == '%') {
24             op2 = pop();
25             op1 = pop();
26             result = op1 + op2;
27             push(result);
28         }
29     }
30     result = pop();
31     printf("Final Result = %d\n", result);
32 }
```

PS C:\Users\ABDUL AZEEZ\Downloads\AT> gcc 2.c -o 2

PS C:\Users\ABDUL AZEEZ\Downloads\AT> ./2

Enter postfix expression (e.g. 32+4*): 32+438*

Push operand: 3 -> Stack top = 3

Push operand: 2 -> Stack top = 2

Final Result = 2

PS C:\Users\ABDUL AZEEZ\Downloads\AT>

EXPLORER

1.c2.c3.c4.c5.c

1.c1.exe2.c2.exe3.c3.exe4.c4.exe5.c5.exe

3.c

```
1 T_val(int, char, int, int)
2 #include <stdio.h>
3
4 int T_eval(int inh_val, char op, int b, int c);
5 int E_eval(int a, int b, int c);
6
7 int T_eval(int inh_val, char op, int b, int c) {
8     int Tval = b;
9     if (op == '*')
10         Tval = Tval * c;
11     printf("T_val computed using inherited base = %d, result = %d\n", inh_val, Tval);
12     return Tval;
13 }
14
15 int E_eval(int a, int b, int c) {
16     int inherited_from_E = a;
17     int Tval = T_eval(inherited_from_E, '*', b, c);
18     int Eval = inherited_from_E + Tval;
19     printf("E_val (final) = %d + %d = %d\n", inherited_from_E, Tval, Eval);
20     return Eval;
21 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

CODE REFERENCE LOG

QUERY RESULTS

powerShell

PS C:\Users\ABDUL AZEEZ\Downloads\AT> gcc 3.c -o 3
>> ./3
Enter values for a, b, c (for expression a + b*c): a + b*c
T_val computed using inherited base = 16, result = 0
E_val (final) = 16 + 0 = 16
Final Result of a + b*c = 16
PS C:\Users\ABDUL AZEEZ\Downloads\AT>

EXPLORER

1.c2.c3.c4.c5.c

1.c1.exe2.c2.exe3.c3.exe4.c4.exe5.c5.exe

4.c

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int isEquivalent(char *t1, char *t2) {
5     return strcmp(t1, t2) == 0;
6 }
7
8 int main() {
9     char type1[20], type2[20];
10
11     printf("Enter first type expression (e.g. int, float, int*): ");
12     scanf("%s", type1);
13     printf("Enter second type expression: ");
14     scanf("%s", type2);
15
16     if (isEquivalent(type1, type2))
17         printf("Types are Equivalent\n");
18     else {
19         printf("Types are NOT Equivalent\n");
20     }
21     char base1[20], base2[20];
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

CODE REFERENCE LOG

QUERY RESULTS

powerShell

PS C:\Users\ABDUL AZEEZ\Downloads\AT> gcc 4.c -o 4
>> ./4
Enter first type expression (e.g. int, float, int*): int , float
Enter second type expression: Types are NOT Equivalent
Reason: Different base types
PS C:\Users\ABDUL AZEEZ\Downloads\AT>

